

PADHAISHURU

GATE Preparation Platform

GATE CH

Chemical Engineering

TREND-BASED MOCK PAPER

— 03 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	10 subjects

Difficulty Distribution

Easy: 17 questions (26%)

Moderate: 38 questions (58%)

Difficult: 10 questions (15%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
These mock papers are constructed using analysis of historical GATE
PYQ patterns.
Historical patterns do not guarantee future examination questions.

GENERAL INSTRUCTIONS

GATE CH — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1. 1M [-0.33]

[General Aptitude]

Simple interest on 1000 at 5% for 2 years:

- (A) 50
- (B) 100
- (C) 200
- (D) 10

Q2. 1M [no -ve]

[Engineering Mathematics]

Partial derivative of x^2y w.r.t x :

Enter your answer here

Q3. 1M [no -ve]

[Process Calculations]

Material balance is based on:

Enter your answer here

Q4. 1M [-0.33]

[Thermodynamics]

Raoult's law for:

- (A) Non-ideal
- (B) Ideal solutions
- (C) Reactions
- (D) None

Q5. 1M [-0.33]

[Fluid Mechanics]

Reynolds number for pipe flow:

- (A) $Re \leq \frac{D \rho v}{4\mu}$
- (B) $Re = \frac{vD}{\nu}$
- (C) Both
- (D) None

Q6. 1M [-0.33]

[Heat Transfer]

Thermal conductivity of metals is:

- (A) High
- (B) Low
- (C) Zero
- (D) Infinite

Q7. 1M [no -ve]

[Mass Transfer]

Fenske equation gives:

Enter your answer here

Q8.

1M [-0.33]

*[Chemical Reaction Engineering]*First order rate: $-r_A = ?$

- (A) $k C_A$
- (B) $k C_A^2$
- (C) k
- (D) k/C_A

Q9.

1M [-0.33]

[Process Control]

Control valve characteristics:

- (A) Linear
- (B) Equal %
- (C) Quick opening
- (D) All

Q10.

1M [-0.33]

[Mechanical Operations]

Terminal velocity in Stokes law:

- (A) $\frac{dp^2(\rho - \rho_A)g}{18\eta}$
- (B) $\frac{dp(\rho - \rho_A)g}{18\eta}$
- (C) $\frac{dp^2}{18\eta}$
- (D) None

Q11.

1M [-0.33]

[Mechanical Operations]

Terminal velocity in Stokes law:

- (A) $\frac{dp^2(\rho - \rho_A)g}{18\eta}$
- (B) $\frac{dp(\rho - \rho_A)g}{18\eta}$
- (C) $\frac{dp^2}{18\eta}$
- (D) None

Q12.

1M [-0.33]

*[Engineering Mathematics]*Partial derivative of x^2y w.r.t x :

- (A) $2xy$
- (B) x^2
- (C) $2x$
- (D) 0

Q13.

1M [-0.33]

[Heat Transfer]

Thermal conductivity of metals is:

- (A) High
- (B) Low
- (C) Zero
- (D) Infinite

Q14.

1M [-0.33]

[Heat Transfer]

Thermal conductivity of metals is:

- (A) High
- (B) Low
- (C) Zero
- (D) Infinite

Q15.

1M [-0.33]

[Mechanical Operations]

Terminal velocity in Stokes law:

- (A) $\frac{dp^2(\rho - \rho')g}{18\eta}$
- (B) $\frac{dp(\rho - \rho')g}{18\eta}$
- (C) $\frac{dp^2}{18\eta}$
- (D) None

Q16.

1M [-0.33]

[Heat Transfer]

Thermal conductivity of metals is:

- (A) High
- (B) Low
- (C) Zero
- (D) Infinite

Q17.

1M [-0.33]

*[Engineering Mathematics]*Partial derivative of x^2y w.r.t x :

- (A) $2xy$
- (B) x^2
- (C) $2x$
- (D) 0

Q18.

1M [-0.33]

[Heat Transfer]

Thermal conductivity of metals is:

- (A) High
- (B) Low
- (C) Zero
- (D) Infinite

Q19.

1M [-0.33]

*[Engineering Mathematics]*Partial derivative of x^2y w.r.t x :

- (A) $2xy$
- (B) x^2
- (C) $2x$
- (D) 0

Q20.

1M [no -ve]

*[Chemical Reaction Engineering]*First order rate: $-r_A = ?$

*Enter your answer here***Q21.**

1M [-0.33]

[Fluid Mechanics]

Reynolds number for pipe flow:

- (A) $Re = \frac{\rho v D}{\mu}$
- (B) $Re = \frac{v D}{\nu}$
- (C) Both
- (D) None

Q22.

1M [-0.33]

[Thermodynamics]

Raoult's law for:

- (A) Non-ideal
- (B) Ideal solutions
- (C) Reactions
- (D) None

Q23.

1M [no -ve]

*[Chemical Reaction Engineering]*First order rate: $-r_A = ?$

*Enter your answer here***Q24.**

1M [-0.33]

[Heat Transfer]

Thermal conductivity of metals is:

- (A) High
- (B) Low
- (C) Zero
- (D) Infinite

Q25.

1M [-0.33]

*[Chemical Reaction Engineering]*First order rate: $-r_A = ?$

- (A) $k C_A$
- (B) $k C_A^2$
- (C) k
- (D) k/C_A

Q26.

1M [no -ve]

[Thermodynamics]

Raoult's law for:

Enter your answer here

Q27.

1M [-0.33]

[Thermodynamics]

Raoult's law for:

- (A) Non-ideal
- (B) Ideal solutions
- (C) Reactions
- (D) None

Q28.

1M [no -ve]

[Heat Transfer]

Thermal conductivity of metals is:

Enter your answer here

Q29.

1M [-0.33]

[Fluid Mechanics]

Reynolds number for pipe flow:

- (A) $Re \leq \frac{D}{\mu}$
- (B) $Re = \frac{vD}{\mu}$
- (C) Both
- (D) None

Q30.

1M [-0.33]

[Mass Transfer]

Fenske equation gives:

- (A) Minimum stages
- (B) Minimum reflux
- (C) Actual stages
- (D) None

Q31.

2M [-0.66]

[Process Calculations]

Material balance is based on:

- (A) $F = ma$
- (B) Conservation of mass
- (C) Energy conservation
- (D) None

Q32.

2M [no -ve]

[Process Calculations]

Material balance is based on:

Enter your answer here

Q33.

2M [-0.66]

[Heat Transfer]

Thermal conductivity of metals is:

- (A) High
- (B) Low
- (C) Zero
- (D) Infinite

Q34.

2M [no -ve]

[Process Calculations]

Material balance is based on:

*Enter your answer here***Q35.**

2M [-0.66]

*[Engineering Mathematics]*Partial derivative of x^2y w.r.t x :

- (A) $2xy$
- (B) x^2
- (C) $2x$
- (D) 0

Q36.

2M [no -ve]

[Fluid Mechanics]

Reynolds number for pipe flow:

*Enter your answer here***Q37.**

2M [no -ve]

[Process Control]

Control valve characteristics:

*Enter your answer here***Q38.**

2M [no -ve]

*[Chemical Reaction Engineering]*First order rate: $-r_A = ?$

*Enter your answer here***Q39.**

2M [-0.66]

[Process Control]

Control valve characteristics:

- (A) Linear
- (B) Equal %
- (C) Quick opening
- (D) All

Q40.

2M [no -ve]

[Process Calculations]

Material balance is based on:

*Enter your answer here***Q41.**

2M [-0.66]

[Process Control]

Control valve characteristics:

- (A) Linear
- (B) Equal %
- (C) Quick opening
- (D) All

Q42.

2M [-0.66]

[Mass Transfer]

Fenske equation gives:

- (A) Minimum stages
- (B) Minimum reflux
- (C) Actual stages
- (D) None

Q43.

2M [no -ve]

[Process Control]

Control valve characteristics:

*Enter your answer here***Q44.**

2M [no -ve]

[General Aptitude]

Simple interest on 1000 at 5% for 2 years:

- (A) 50
- (B) 100
- (C) 200
- (D) 10

Q45.

2M [-0.66]

[Process Control]

Control valve characteristics:

- (A) Linear
- (B) Equal %
- (C) Quick opening
- (D) All

Q46.

2M [-0.66]

[General Aptitude]

Simple interest on 1000 at 5% for 2 years:

- (A) 50
- (B) 100
- (C) 200
- (D) 10

Q47.

2M [no -ve]

[Process Calculations]

Material balance is based on:

*Enter your answer here***Q48.**

2M [-0.66]

[General Aptitude]

Simple interest on 1000 at 5% for 2 years:

- (A) 50
- (B) 100
- (C) 200
- (D) 10

Q49.

2M [no -ve]

[Thermodynamics]

Raoult's law for:

Enter your answer here

Q50.

2M [-0.66]

[Fluid Mechanics]

Reynolds number for pipe flow:

- (A) $Re \leq \frac{D \rho v}{4\mu}$
- (B) $Re = \frac{vD}{\nu}$
- (C) Both
- (D) None

Q51.

2M [-0.66]

[Mass Transfer]

Fenske equation gives:

- (A) Minimum stages
- (B) Minimum reflux
- (C) Actual stages
- (D) None

Q52.

2M [-0.66]

[General Aptitude]

Simple interest on 1000 at 5% for 2 years:

- (A) 50
- (B) 100
- (C) 200
- (D) 10

Q53.

2M [no -ve]

*[Engineering Mathematics]*Partial derivative of x^2y w.r.t x :

Enter your answer here

Q54.

2M [-0.66]

[Process Control]

Control valve characteristics:

- (A) Linear
- (B) Equal %
- (C) Quick opening
- (D) All

Q55.

2M [no -ve]

[Heat Transfer]

Thermal conductivity of metals is:

- (A) High
- (B) Low
- (C) Zero
- (D) Infinite

Q56.

2M [no -ve]

*[Chemical Reaction Engineering]*First order rate: $-r_A = ?$

*Enter your answer here***Q57.**

2M [-0.66]

[Fluid Mechanics]

Reynolds number for pipe flow:

- (A) $Re = \frac{\rho v D}{\mu}$
- (B) $Re = \frac{v D}{\nu}$
- (C) Both
- (D) None

Q58.

2M [-0.66]

[Process Control]

Control valve characteristics:

- (A) Linear
- (B) Equal %
- (C) Quick opening
- (D) All

Q59.

2M [-0.66]

[Heat Transfer]

Thermal conductivity of metals is:

- (A) High
- (B) Low
- (C) Zero
- (D) Infinite

Q60.

2M [-0.66]

[General Aptitude]

Simple interest on 1000 at 5% for 2 years:

- (A) 50
- (B) 100
- (C) 200
- (D) 10

Q61.

2M [-0.66]

[Process Control]

Control valve characteristics:

- (A) Linear
- (B) Equal %
- (C) Quick opening
- (D) All

Q62.

2M [no -ve]

[Fluid Mechanics]

Reynolds number for pipe flow:

- (A) $Re = \frac{\rho v D}{\mu}$
- (B) $Re = \frac{v D}{\nu}$
- (C) Both
- (D) None

Q63.

2M [-0.66]

[Thermodynamics]

Raoult's law for:

- (A) Non-ideal
- (B) Ideal solutions
- (C) Reactions
- (D) None

Q64.

2M [-0.66]

*[Engineering Mathematics]*Partial derivative of x^2y w.r.t x :

- (A) $2xy$
- (B) x^2
- (C) $2x$
- (D) 0

Q65.

2M [no -ve]

[Mechanical Operations]

Terminal velocity in Stokes law:

Enter your answer here